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Field Crops Research 89 (2004) 369–378
Field Crops Research
www.elsevier.com/locate/fcr

# Comparison of two methods for estimating parameters of harvest index increase during seed growth

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Received 17 October 2003; received in revised form 19 February 2004; accepted 22 February 2004

**Abstract**

The linearity of harvest index (HI) increase has provided a simple means to analyze and predict seed growth and yield in experimental and simulation studies. When using the linear HI concept, it is necessary to estimate three parameters: (1) the period from anthesis to the onset of linear increase in HI (LAG), (2) the time of cessation of linear increase in HI (HIM), and (3) the rate of linear increase in HI with time (dHI/d*t*). Thus, an appropriate method is needed for accurate estimation of the parameters. In this paper, we compare the capability of two methods for this purpose. In the first, conventional method (M1), a simple, linear regression model ($y = a + bx$) is used to describe HI increase versus time using data contained within the main seed growth period. dHI/d*t* is taken as *b* and LAG as $-a/b$. In this method, physiological maturity is considered as HIM. The second method (M2) is based on applying a non-linear, segmented regression model. The segmented model consists of two intersecting lines, a sloping line ($y = a + bx$) for the linear increase in HI ($b = \mathrm{dHI}/\mathrm{d}t$) and LAG ($-a/b$) and a horizontal line ($y = a + bx_0$) which determines HIM ($x_0$) and consequently maximum HI ($\mathrm{HI}_{max}$). Data sets of HI versus time for 13 spring wheat (*Triticum aestivum* L.) cultivars grown under rainfed conditions were used. The segmented model described HI increase with time well and its performance was comparable to quadratic, cubic and logistic models that have been used for this purpose. $R^2$ values for the segmented model were between 0.93 and 0.98. M1 and M2 provided considerably different estimates of LAG, HIM and dHI/d*t*. Comparison of measured maximum HI with those calculated using parameter estimates of the conventional method (M1) showed that this method is not accurate (RMSE = 5.4%). The deficiency in M1 in estimating HI-related parameters may be a source of error in grain yield prediction based on the linear HI increase concept. M2, however, was significantly better (RMSE = 0.3%) and thus is proposed as the preferred method for future use.


*Keywords:* Harvest index; Yield prediction; Parameter estimation; Simulation model; Wheat

## 1. Introduction

Crop models have proved to be useful tools in increasing quantitative understanding of how crop systems operate (Sinclair and Seligman, 2000). The heuristic value of crop models for a number of purposes is well established (Sinclair and Seligman, 1996). The main goal of many crop models is to predict crop yield, although they have also been used for prediction of processes such as development, growth and water use. Many models predict seed yield from estimates of the number of seeds per unit area and the rate and duration of seed filling (e.g. Ritchie et al., 1998). Some other models use partitioning coefficients to estimate seed yield accumulated (e.g. Goudriaan and van Laar, 1994).

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0378-4290/$ – see front matter 
doi:10.1016/j.fcr.2004.02.015

It has been shown that the prediction of seed number is a source of error in crop models (Porter et al., 1993). Moot et al. (1996) stated that a simpler description might avoid these problems.

Speath and Sinclair (1985) found in soybean that seed growth might be better defined by determining the change in harvest index (HI) with time during the seed growth period. They found that HI increased linearly with time (dHI/d$t$) over much of the seed growth period. They observed this linearity in HI increase with time across three locations and a total of 18 cultivars. This response has also been observed in other crops such as sorghum and maize (Muchow, 1988), peanut (Bennett et al., 1993), wheat (Moot et al., 1996), field pea (Lecoeur and Sinclair, 2001) and pigeonpea (Robertson et al., 2001a). It has also been shown that dHI/d$t$ remains stable over a range of growth conditions such as variations in sowing date, irrigation treatments and nitrogen level (Moot et al., 1996; Bange et al., 1998; Bindi et al., 1999; Lecoeur and Sinclair, 2001). However, Bange et al. (1998) in sunflower and Hammer and Broad (2003) in sorghum found that dHI/d$t$ was reduced considerably under cool conditions.

Using a constant dHI/d$t$ means that it is necessary only to track total crop mass and multiply the mass by the dHI/d$t$ to estimate seed growth, rather than developing a series of equations to describe various aspects of seed development and growth (Bindi et al., 1999). If desired this approach can be used to calculate the daily seed growth rate (Sinclair, 1986). The strength of this approach lies in its simplicity, intrinsically combining the contribution of current and stored assimilate to seed yield and so removing the need for complex predictions of seed number and size in the prediction of seed yield (Chapman et al., 1993). Use of a constant dHI/d$t$ has proved effective and robust in a number of crop simulation models including soybean (Sinclair, 1986), maize (Muchow et al., 1990), wheat (Amir and Sinclair, 1991), sunflower (Chapman et al., 1993), peanut (Hammer et al., 1995), chickpea (Soltani et al., 1999) and pigeonpea (Robertson et al., 2001b). However, Hammer and Muchow (1994) showed that the lower precision in predicting grain yield in their sorghum model was associated with limitation in predicting HI.

When using the linear HI increase concept to predict seed yield in crop simulation models, it is necessary to estimate three parameters for these models. These are:

1. lag phase duration (LAG), i.e., the period from anthesis to the onset of linear HI increase;
2. the time of cessation of the linear increase in HI (HIM), i.e., the time at which maximum HI occurs; and
3. the rate of linear increase in HI with time (dHI/d$t$).

These three parameters define the maximum attainable HI ($HI_{max}$). This parameter is sometimes used in preference to the time of cessation of linear HI increase. For example, Hammer and Muchow (1994) used a maximum HI of 55% to reflect the genetic potential of most current commercial sorghum cultivars.

Small variation in rate of linear increase in HI and the time of start and cessation of linear increase have major effects on yield prediction (Hammer and Muchow, 1994; Hammer and Broad, 2003). Grain yield prediction is very sensitive to the changes of HI-related parameters. Sinclair (1986) in sensitivity test of his soybean model found that grain yield changed by −24 and 8% by 20 and −20% change in dHI/d$t$, respectively. Similarly, Chapman et al. (1993) in sensitivity analysis of their sunflower model showed that −20 and 20% change in dHI/d$t$ caused −21 to 20% change in grain yield prediction under different conditions with respect to water availability. Therefore, it can be concluded that accurate estimation of HI parameters are needed for precise estimation of crop yield by crop simulation models. Inaccurate estimation of HI parameters is likely a source of error in yield prediction by crop models. On the other hand, HI-related parameters should be estimated for new species or cultivars. Also, modified and refined estimates are required, as new data sets become available.

To obtain HI parameters, ordinarily a simple, linear regression model ($y = a + bx$) is used to describe HI increase versus time (e.g. Moot et al., 1996) or thermal time (e.g. Lecoeur and Sinclair, 2001) after deletion of data from either early or late stages of seed growth. Then, dHI/d$t$ is taken as $b$ and the time of the onset of linear increase as $-a/b$. Unfortunately, in the majority of the references that have used HI approach for yield prediction, there is not clear description of how HIM was determined. Generally, this parameter has been determined independently from other HI parameters in a qualitative (not quantitative) manner. This parameter has been determined by visual inspection of the plot of HI versus time (Salado-Navaro et al., 1985; Chapman et al., 1993; Hammer and Muchow, 1994) or by occurrence of physiological maturity, which is determined using visual symptoms (Muchow et al., 1990; Amir and Sinclair, 1991; Hammer et al., 1995). In some situations, data are only available from the middle stages of the seed growth period, so only a simple, linear regression model can be used. Hereafter, we denote this conventional approach as M1.

In another approach, Hammer and Broad (2003) used a two-phase segmented regression model to describe HI increase versus time during seed growth. This model separates HI increase into distinct phases; linear increase in HI occurs in phase 1 and maximum

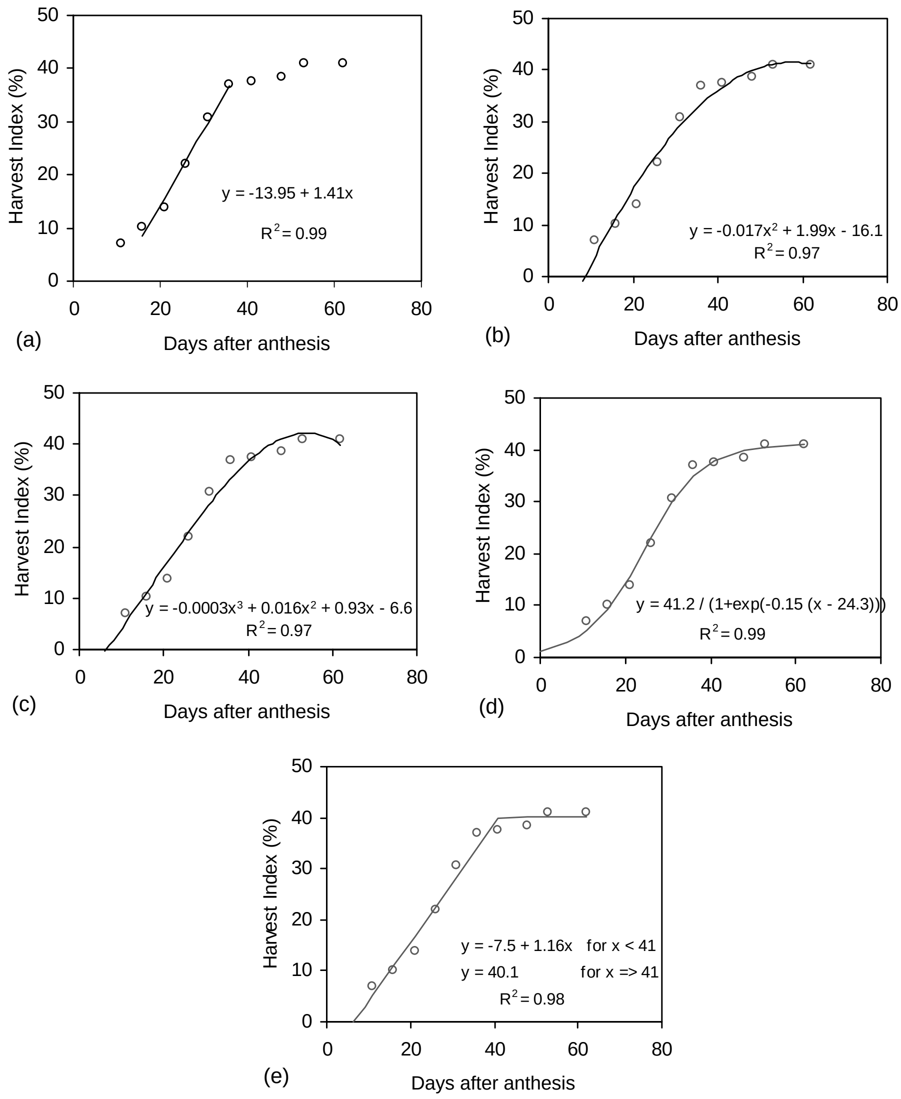


Fig. 1. Harvest index (%) vs. time for cv. Inia in replication 1. The solid line is linear (a), quadratic (b), cubic (c), logistic (d) or segmented (e) regression model fitted to the data.

HI in phase 2. Mathematically, the segmented model may be expressed as (Fig. 1e):

$$y = \begin{cases} a + bx & \text{for } x < x_0, \\ a + bx_0 & \text{for } x > x_0 \end{cases} \tag{1}$$

where $y$ is the harvest index (%), $x$ the time after anthesis (day), $a$ the intercept with the vertical axis ($x = 0$, %), $b$ the rate of linear increase in HI (dHI/d$t$, % per day), $x_0$ the time of cessation of the linear increase in HI. We call $x_0$ harvest index maturity (HIM), analogous to physiological maturity. From these parameters three further parameters can be obtained as:

| | |
|---|---|
| $-a/b$ | duration of lag phase (LAG, day) |
| HIM−LAG | duration of linear phase (LPD, day) |
| $a + bx_0$ | maximum harvest index ($HI_{max}$, %) |

While the approach of Hammer and Broad (2003) seems interesting, we could not find any other application of this method. In one study, Soltani et al. (1999) used such a model to estimate the time of start and end of the period of linear HI increase in chickpea, but they used the data of grain weight versus time not HI versus time. Hereafter, we denote this approach as M2.

So far no evaluation and/or comparison have been made between M1 and M2 methods with respect to their ability and precision in estimating HI parameters. Therefore, our objectives of this study were: (1) to evaluate and compare capability of the two methods outlined for estimating parameters of HI increase during seed growth, and (2) to indicate how the refined estimates of HI parameters can be used to explore the heritability of traits associated with the HI concept.

## 2. Materials and methods

### 2.1. Data sets

Data sets for statistical analyses came from a field experiment on 13 spring wheat cultivars. The field experiment was conducted at the Gorgan University of Agricultural Sciences Research Farm, Gorgan (36°85′N, 54°27′E and 100 m asl) in the growing season of 1998–1999 under rainfed conditions. The soil was a deep silty clay. The 13 cultivars were sown in a randomized complete block design with four replications. Plots (2 m wide × 5 m long) were hand-seeded on 9 December using row spacing of 20 cm. Seeding rate was calculated for each cultivar using percentage germination and 1000-seed weight to achieve a density of 300 plant m$^{-2}$. Prior to seeding 150 kg ha$^{-1}$ superphosphate was broadcasted and incorporated into the soil. Plots were top-dressed at tillering, stem elongation and heading stages with urea (each 50 kg ha$^{-1}$). Weeds were hand-controlled. During the seed filling period, daily mean air temperatures varied between 12 and 26 °C.

Plant samples were taken from each plot at 5-day intervals commencing 7 days after anthesis. The number of samplings was 10–11. Sampling involved taking 20 plants and then selecting five plants for measurement. Seeds were separated from leaves, stems and ears, before all plant parts were oven-dried at 70 °C to constant weight, and then weighed. HI was determined based on above-ground plant mass as the ratio of seed weight to total weight of the harvested material. This calculation was carried out for each treatment and replication at each harvest time.

### 2.2. Analysis

In M1, a simple, linear regression model was fitted to data of HI versus time. Data points were removed successively from early and/or late stages of seed growth until the greatest $R^2$ and minimum RMSE were obtained. This was tedious as many fits were required. Recorded time of physiological maturity was taken as HIM. In M2, the HI data were regressed against time (days after anthesis) using a two-piece segmented, non-linear regression model (Eq. (1)). The segmented model was fitted to data sets with the multivariate secant (do not use derivates, DUD) option of the non-linear regression (NLIN) procedure in SAS Institute (1989). The DUD option of the NLIN procedure for non-linear regression in SAS provides an easy method for fitting various non-linear models to HI data, in that partial derivatives for the parameters are not required. Only initial parameter estimates are needed to seed the iterative procedure.

In addition to the simple, linear and the segmented models, regressions were performed using quadratic,

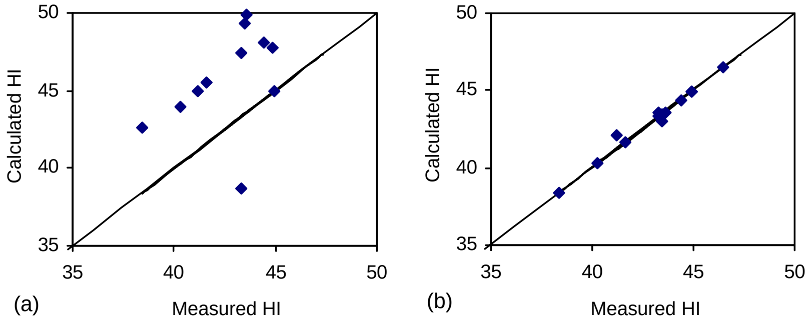


Fig. 2. Comparison of maximum measured HI with values calculated using M1 (a) and M2 (b). The solid line is 1:1 line.

cubic and logistic models to describe HI as a function of time. The logistic model was (Fig. 2d):

$$y = \frac{c}{1 + \exp[-a(x - b)]} \quad (2)$$

where $c$ is the maximum HI (%), $b$ the time when HI reaches half its maximum value and $a$ the steepness of increase in HI. This model was also fitted to data sets using the NLIN procedure in SAS. Quadratic and cubic models were fitted using the general linear model (GLM) procedure in SAS. The adequacy of fit of the selected models was compared by computing $R^2$ and root mean square error (RMSE). The regression analysis was done using combined data of each cultivar cover replications (i.e., 13 data sets).

## 3. Results and discussion

### 3.1. Statistical comparison of the selected regression models

An example of the fit of the selected regression models to data of HI versus time after anthesis is shown in Fig. 1. The selected models described well the increase in HI over time as indicated by the high $R^2$ values of 0.99 for linear, 0.97 for quadratic and cubic, 0.99 for logistic and 0.98 for segmented models. Increase in HI followed a sigmoid pattern (Fig. 1d). Consequently, the logistic model provided the best fit, although this advantage was not statistically significant compared to other models (data not shown). The segmented model (M2) presented a successful approximation of increase in HI. According to this model the linear phase of increase in HI began 6.5 days after anthesis, proceeded with a rate (dHI/d$t$) of 1.16% per day and terminated after 41 days when HI reached a maximum of 40%. Best fit of the linear model (M1) was achieved with HI data from 16 to 36 days after anthesis as $y = -13.95 + 1.41x$; $n = 5$, RMSE $=$ 1.51, $R^2 = 0.99$. The resultant dHI/d$t$ of 1.41% per day and lag phase duration of 9.9 days were significantly different to values obtained with the segmented model. This deviation may lead to bias in seed yield prediction, especially if observed physiological maturity (44 days after anthesis for this cultivar) was used as the time of cessation of linear increase in HI.

Comparison of RMSE and $R^2$ from the selected models showed that the segmented model had an equal or smaller RMSE than linear, quadratic, cubic and logistic models in 8, 11, 9 and 6 cases out of the 13 cases, respectively (Table 1). In other cases, RMSE from the segmented model was greater, but the difference was small and not statistically significant (data not shown). With respect to $R^2$, the segmented model had a greater or equal $R^2$ value than linear, quadratic, cubic and logistic models in 12, 13, 12 and 1 case of the 13 cases, respectively (Table 1). When the segmented model indicated a lower $R^2$, the magnitude of this deficit was less than 2% (0.99 versus 0.97 for cv. Pastor) and was not statistically significant (data not shown).

Therefore, it can be concluded that all the selected models are similar for describing increase in HI during the seed growth period from a statistical point of view. The segmented model successfully approximates this increase and has parameters that are well defined for modeling purposes or analysis of experimental treatments.

Table 1
Comparison of linear (L), quadratic (Q), cubic (C), logistic (G) and segmented (S) regression models to describe increase in harvest index during seed growth for 13 spring wheat cultivars. Root mean square error and coefficient of determination ($R^2$) are included

| Cultivar | RMSE | | | | | $R^2$ | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | L | Q | C | G | S | L | Q | C | G | S |
| Alborz | 2.93 | 3.24 | 3.18 | 1.96 | 3.48 | 0.94 | 0.97 | 0.97 | 0.99 | 0.98 |
| Atila | 3.23 | 3.72 | 3.49 | 2.85 | 2.88 | 0.94 | 0.96 | 0.96 | 0.97 | 0.97 |
| Atrak | 2.89 | 3.67 | 3.27 | 2.64 | 2.92 | 0.94 | 0.95 | 0.96 | 0.97 | 0.96 |
| Bakonara | 2.01 | 2.71 | 2.30 | 1.28 | 2.82 | 0.98 | 0.97 | 0.98 | 0.99 | 0.98 |
| Falat | 2.14 | 3.79 | 3.20 | 2.72 | 2.90 | 0.97 | 0.95 | 0.96 | 0.97 | 0.96 |
| Golestan | 2.37 | 3.79 | 3.15 | 2.54 | 2.81 | 0.96 | 0.95 | 0.97 | 0.98 | 0.97 |
| Inia | 4.60 | 4.38 | 4.40 | 4.08 | 2.61 | 0.87 | 0.92 | 0.93 | 0.93 | 0.93 |
| Khazar 1 | 2.53 | 2.78 | 2.80 | 2.45 | 2.45 | 0.95 | 0.96 | 0.96 | 0.97 | 0.96 |
| Pastor | 3.02 | 4.24 | 2.54 | 2.22 | 3.63 | 0.96 | 0.95 | 0.98 | 0.99 | 0.97 |
| P.R.1 | 3.08 | 3.97 | 3.29 | 2.72 | 2.66 | 0.93 | 0.95 | 0.97 | 0.98 | 0.97 |
| Tajan | 3.89 | 3.78 | 3.80 | 3.35 | 2.03 | 0.92 | 0.94 | 0.94 | 0.95 | 0.94 |
| Viniak | 3.32 | 3.63 | 3.60 | 2.84 | 2.72 | 0.93 | 0.94 | 0.94 | 0.96 | 0.95 |
| Zagros | 3.09 | 3.94 | 3.91 | 3.24 | 1.80 | 0.94 | 0.95 | 0.95 | 0.97 | 0.96 |

### 3.2. Comparison of the estimation methods

Estimates of the parameters resulting from fits using M1 and M2 are shown in Table 2. Corresponding RMSE and $R^2$ values can be found in Table 1. dHI/d$t$ ranged between 1.16 and 1.52% per day for M1, and between 1.06 and 1.39% per day for M2. These ranges are narrower than the 0.64–1.37% per day reported by Moot et al. (1996) for wheat. However, average dHI/d$t$ of 1.35% per day for M1 and 1.25% per day for M2 were higher than the 1.00% per day reported by Moot et al. (1996). For 11 of the 13 cultivars, M1 resulted in greater dHI/d$t$ than M2, but for 2 of 13 cultivars (Zagros and Inia) the estimates were the same (Table 2).

Table 2
Parameter estimates using M1 and M2 for 13 spring wheat cultivars. Parameter estimates are $a$: intercept (%), dHI/d$t$: linear increase in harvest index (% per day), LAG: duration of lag phase (day), HIM: harvest index maturity (days after anthesis), and LPD: linear phase duration (day)

| Cultivar | $a$ | | dHI/d$t$ | | LAG | | HIM | | LPD | |
|---|---|---|---|---|---|---|---|---|---|---|
| | M1 | M2 | M1 | M2 | M1 | M2 | M1 | M2 | M1 | M2 |
| Alborz | −10.26 | −7.09 | 1.518 | 1.395 | 6.8 | 5.1 | 42.5 | 35.9 | 35.7 | 30.8 |
| Atila | −7.32 | −6.06 | 1.359 | 1.312 | 5.4 | 4.6 | 40.8 | 38.5 | 35.4 | 33.8 |
| Atrak | −11.42 | −6.46 | 1.511 | 1.329 | 7.6 | 4.9 | 37.3 | 38.6 | 29.7 | 33.8 |
| Bakonara | −7.64 | −6.14 | 1.403 | 1.347 | 5.4 | 4.6 | 39.5 | 37.9 | 34.1 | 33.3 |
| Falat | −9.46 | −4.81 | 1.421 | 1.229 | 6.7 | 3.9 | 40.0 | 39.2 | 33.3 | 35.2 |
| Golestan | −9.87 | −5.17 | 1.380 | 1.198 | 7.1 | 4.3 | 39.0 | 38.0 | 31.9 | 33.6 |
| Inia | −4.70 | −4.70 | 1.167 | 1.167 | 4.0 | 4.0 | 43.0 | 39.7 | 39.0 | 35.6 |
| Khazar 1 | −3.60 | −2.20 | 1.290 | 1.198 | 2.8 | 1.8 | 41.0 | 38.1 | 38.2 | 36.3 |
| Pastor | −7.54 | −5.23 | 1.200 | 1.151 | 6.3 | 4.5 | 38.5 | 42.4 | 32.2 | 37.9 |
| P.R.1 | −8.16 | −2.85 | 1.317 | 1.064 | 6.2 | 2.7 | 40.3 | 42.3 | 34.1 | 39.6 |
| Tajan | −7.96 | −6.45 | 1.472 | 1.375 | 5.4 | 4.7 | 41.3 | 38.5 | 35.9 | 33.8 |
| Viniak | −10.05 | −7.84 | 1.394 | 1.317 | 7.2 | 6.0 | 43.0 | 39.1 | 35.8 | 33.1 |
| Zagros | −5.23 | −5.24 | 1.158 | 1.158 | 4.5 | 4.5 | 41.3 | 37.7 | 36.8 | 33.2 |
| Mean | −7.94** | −5.40 | 1.353** | 1.249 | 5.8** | 4.3 | 40.6* | 38.9 | 34.8 | 34.6 |

* Significant at 0.05 level of probability based on paired $t$-test.
** Significant at 0.01 level of probability based on paired $t$-test.

The difference between M1 and M2 was significant at the 0.01 level with respect to dHI/d$t$. Lag phase duration varied between 3 and 8 days for M1 and between 2 and 6 days for M2, both considerably less than the range reported by Moot et al. (1996). However, as lag phase duration is controlled by temperature (Sofield et al., 1977; Bange et al., 1998), this may explain the differences between experiments. The larger estimates of lag phase duration obtained with M1 differed significantly ($P = 0.01$) from the estimates obtained with M2. Duration of linear increase phase ranged between 30 and 39 days for M1 and between 31 and 40 days for M2. Longer estimates of harvest index maturity using M1 were cancelled by larger estimates of lag phase duration, therefore, there was no significant difference between the methods with respect to linear phase duration (Table 2). Harvest index maturity (from M2) occurred 36–42 days after anthesis. However, physiological maturity, HIM for M1, ranged from 37 to 43 days after anthesis. Therefore, the cessation of linear increase in HI occurred earlier than physiological maturity (Table 2). This difference was statistically significant ($P = 0.05$). Thus, using physiological maturity as the time of cessation of linear increase in HI might result in bias when predicting seed yield using crop simulation models. Lag and linear phase duration, and harvest index maturity found in the present study are not suitable to be used in crop models, because they have obtained under rainfed (water limited) conditions. Here, our objective was not to obtain valid HI parameters for modeling purposes.

Measured maximum HI values ranged between 38 (cv. Zagros) and 47% (cv. Tajan), with mean 43%. This relatively low maximum HI was a consequence of the rainfed conditions in this study. Fig. 2 compares measured maximum HI with values calculated using M1 and M2. Calculated maximum HI was obtained by multiplying linear phase duration and dHI/d$t$. M1 generally over-estimated maximum HI (Fig. 2a). For 11 of the 13 cultivars, calculated maximum HI by M1 was greater than that observed, for 1 cultivar a large underestimation was observed (38.7% versus 43.3%), and for 1 cultivar calculated HI was the same as measured HI. Over-estimation of maximum HI using M1 was due to over-estimation of dHI/d$t$ using this method, because both the methods were similar with respect to linear phase duration. RMSE of prediction of HI for M1 was 5.4%. M2, on the other hand, was very successful; calculated HIs were scattered close to the 1:1 line with RMSE of 0.3% (Fig. 2b).

Table 3
Mean values of duration of lag phase (LAG, day), linear phase duration (LPD, day), harvest index maturity (HIM, days after anthesis), linear increase in harvest index (dHI/d$t$, % per day) and maximum harvest index ($HI_{max}$, %) in 13 spring wheat cultivars. $F$ statistics and their significance are also included

| Cultivar | LAG | LPD | HIM | dHI/d$t$ | $HI_{max}$ |
|---|---|---|---|---|---|
| Alborz | 5.1 a[a] | 31.1 d | 36.2 c | 1.385 ab | 43.1 abc |
| Atila | 4.5 ab | 34.2 bcd | 38.7 abc | 1.300 abc | 44.3 ab |
| Atrak | 4.8 ab | 34.2 bcd | 39.1 abc | 1.314 abc | 44.9 ab |
| Bakonara | 4.8 ab | 32.5 bcd | 37.4 bc | 1.387 ab | 45.7 ab |
| Falat | 4.0 abc | 34.8 abc | 38.8 abc | 1.248 bc | 43.6 abc |
| Golestan | 4.4 ab | 33.6 bcd | 38.0 bc | 1.214 bc | 41.0 bc |
| Inia | 4.4 ab | 35.1 abc | 39.4 abc | 1.188 bc | 42.1 abc |
| Khazar 1 | 2.0 c | 35.6 ab | 37.6 bc | 1.224 bc | 43.8 abc |
| Pastor | 4.6 ab | 37.9 a | 42.5 a | 1.155 c | 43.8 abc |
| P.R.1 | 2.9 bc | 38.2 a | 41.1 ab | 1.102 c | 41.1 abc |
| Tajan | 5.5 a | 31.5 cd | 36.9 c | 1.494 a | 48.7 a |
| Viniak | 6.0 a | 33.4 bcd | 39.4 abc | 1.311 abc | 43.8 abc |
| Zagros | 4.6 ab | 32.9 bcd | 37.5 bc | 1.175 bc | 38.5 c |
| Mean | 4.4 | 34.2 | 38.7 | 1.269 | 43.1 |
| $F$-value | 2.06* | 2.87** | 1.53 | 2.20* | 1.29 |

[a] Means with a common letter in each column are not significantly different based on least significant difference (LSD) test.
* Significant at 0.05 level of probability.
** Significant at 0.01 level of probability.

Therefore, it can be concluded that the conventional method, M1, for estimation of HI parameters is not an accurate method. M1 has also several insufficiencies as: (1) using this method is tedious, (2) only the data from middle part of seed growth period are used, (3) all the parameters are not obtain simultaneously and by statistical methods such as least-squares method, and (4) the cessation time of linear increase in HI and physiological maturity may occur sequentially not synchronously. The deficiency of the M1 in estimating HI-related parameters will likely be a source of error in grain yield prediction based on the linear HI increase concept.

### *3.3. Using M2 in experimental analysis*

M2 can be used to estimate HI-related parameters when repeated estimates are needed for further analyses to interpret treatment effects. To indicate such an application we examined the phenotypic and genetic coefficients of variations (PCV and GCV), broad-sense heritability and genetic advance of the HI parameters in the 13 spring wheat cultivars (Table 4). To do this, the segmented model was separately fitted to data of each cultivar for each replication ($13 \times 4$ data sets) so four estimates were obtained for each parameter. These estimates were then subjected to variance and covariance analyses based on randomized complete block design. The variance components, genotypic coefficient of variation, broad-sense heritability, and estimated genetic advance were determined as suggested by Burton and DeVane (1953) and Johnson et al. (1955).

Mean values of HI parameters obtained in this way are not exactly equal to those obtained using pooled data, but the differences are small and not statistically significant (Table 3 versus Table 2). Analysis of variance (*F*-test) showed that genotypic difference for lag phase duration, linear phase duration and dHI/d*t* were significant, but not for harvest index maturity or $HI_{max}$ (Table 3). Moot et al. (1996), examining dHI/d*t* and lag phase in wheat cultivars from five experiments, stated that varietal differences can be further studied to determine whether the varietal differences are genetically controlled and whether they are amenable to direct selection for high dHI/d*t* values or shorter lag phase.

The PCV was generally higher than the GCV for all the HI parameters. The highest value for GCV was observed for lag phase length. Linear phase duration,

Table 4
Phenotypic (PCV) and genetic coefficient of variation (GCV), broad-sense heritability (*H*, %), expected genetic advance (GA) and expected genetic advance as a percent of mean ($GA_m$, %) for parameters describing increase in harvest index during seed growth in 13 spring wheat cultivars

| Parameter | PCV | GCV | *H* | Ga[a] | $GA_m$ |
|---|---|---|---|---|---|
| Lag phase duration (day) | 23.3 | 15.7 | 45.6 | 0.8 | 19.2 |
| Linear phase duration (day) | 6.3 | 5.1 | 65.2 | 2.5 | 7.2 |
| Harvest index maturity (day) | 4.5 | 2.6 | 34.5 | 1.1 | 2.7 |
| Linear increase in harvest index (% per day) | 8.7 | 6.1 | 49.5 | 0.1 | 7.5 |
| Maximum harvest index (%) | 4.9 | 2.4 | 22.7 | 0.9 | 2.0 |

[a] The selection differential used was 1.755, corresponding to a selection intensity of 10%.

dHI/d*t*, harvest index maturity and $HI_{max}$ had small GCVs (Table 4). Moderate estimates of heritability were found for lag phase and linear phase duration and dHI/d*t*, indicating considerable genetic control on these traits. Low estimates were obtained for harvest index maturity and $HI_{max}$, suggesting that environmental effects constitute a considerable portion of the total phenotypic variation in this population (Table 4). The expected genetic advance based on the selection of 10% of genotypes, expressed as a percentage of the mean, ranged from 2% for $HI_{max}$ to 19% for lag phase. Except for the length of lag phase, genetic advance estimates were moderate or low. Moderate to low estimates of GCV, heritability and genetic advance for HI parameters indicate that introducing novel genetic material and conducting crossing programs may be necessary for improvement of these traits.

Correlation coefficients among HI-related parameters are shown in Table 5. Maximum HI showed

Table 5
Correlation coefficients among HI-related parameters in 13 spring wheat cultivars. The parameters are the duration of lag phase (LAG), linear phase duration (LPD), harvest index maturity, linear increase in harvest index (dHI/d*t*) and maximum HI ($HI_{max}$)

| | LAG | LPD | HIM | dHI/d*t* | $HI_{max}$ |
|---|---|---|---|---|---|
| LAG | 1.00 | | | | |
| LPD | −0.34* | 1.00 | | | |
| HIM | 0.22 | 0.85** | 1.00 | | |
| dHI/d*t* | 0.37 | −0.72** | −0.53** | 1.00 | |
| $HI_{max}$ | 0.23 | −0.10 | 0.03 | 0.76** | 1.00 |

* Significant at 0.05 level of probability.
** Significant at 0.01 level of probability.

a highly positive correlation with dHI/d*t*, but there was not correlation between maximum HI and other HI parameters, i.e., lag and linear phase duration and harvest index maturity. However, a significant negative correlation was found between dHI/d*t* and linear phase duration and harvest index maturity. This negative correlation may reduce gain from a higher dHI/d*t*. More studies should be conducted to determine whether this negative correlation are due to genetic linkage. If yes, breeding efforts aimed at improving dHI/d*t* and hence maximum HI would have to find genetic exception to this negative relation.

## 4. Conclusions

The main objective of this paper was to compare two methods used for estimating HI parameters. We found that:

(1) The conventional method, M1, for estimation of HI parameters is not an accurate method. The deficiency of the M1 in estimating HI-related parameters may be a source of error in grain yield prediction based on the linear HI increase concept. This method should be abandoned and M2 should be used instead.
(2) The segmented model used in M2 describes HI increase with time well and the performance of the model was comparable to quadratic, cubic and logistic models that have been used for this purpose.
(3) With the segmented model of M2, it is possible to obtain non-biased, simultaneous estimates of all HI parameters without need to remove of data from early and/or late stages of seed growth.
(4) M2 can be applied to estimate each parameter when only one estimate is required, e.g. for incorporation in a crop model. Alternatively, the model may be applied to estimate HI parameters separately from data of each replication. These estimates can then be used to analyze effect of experimental treatments on HI parameters.
(5) For practical situations, the best way to estimate HI parameters is to collect HI data during seed growth period and after physiological maturity, and then fitting the segmented model of M2 to obtain the parameters.

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